hCRI.io

The Complete Guide

Understanding and using hCRI.io for LED color-rendering analysis — CRI, TM-30, spectral power distribution, and everything the report page shows you.

LED · TM-30 · CIE 13.3-1995 CRI · Color Rendering
hcri.io

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1. What is hCRI.io?

hCRI.io is a web application that turns a spectral measurement of a light source into a plain-English report on how well that light renders color. Every lamp, LED, or flashlight reproduces color a little differently — hCRI.io reads the light's spectral power distribution (SPD) and computes the industry-standard metrics used to describe that behavior: CRI (Ra and R1–R15), the newer IES TM-30-18 Rf/Rg system, correlated color temperature (CCT), tint (Duv) and luminous efficacy (LER).

You upload a spectral measurement (or a compatible PDF export from a spectrometer), and hCRI.io produces a full report: overall fidelity and gamut scores, a breakdown by hue region, a spectral power distribution chart, and a chromaticity plot — plus a plain-language written analysis and a preview of how the light would render a photograph.

Who it's for

- **Lighting designers and specifiers** comparing fixtures or verifying a manufacturer's claimed CRI/TM-30 numbers.
- **Photographers and videographers** checking whether a light will render skin tones and colors accurately.
- **Hobbyists and flashlight enthusiasts** who want more than a single CRI number on the box.
- **Anyone with a spectrometer** (or access to one) who wants a fast, shareable, plain-language readout.

What you get from a report

- TM-30 Rf (fidelity) and Rg (gamut) scores, plus the classic CRI Ra and R9 (deep red) scores.
- A 16-bin breakdown showing exactly which hue ranges shift, and by how much.
- A 99-sample breakdown, for a finer-grained look than the 16-bin view alone.
- The raw spectral power distribution, plotted against the reference illuminant.
- CCT and Duv — how warm/cool the light is, and how far it sits from the pure blackbody line.
- LER, the luminous efficacy of the radiation, with a chart comparing your spectrum to the eye's sensitivity.
- A downloadable IES TM-30-18 report page (PNG or PDF) suitable for spec sheets and submittals.
- A license on every report, and an optional raw-data download, both under your control.

2. Why color rendering matters

Two lights can share the same color temperature and still render color very differently. A 3000K LED and a 3000K incandescent bulb both look “warm white” to the eye, but the LED may be emitting a spiky, narrow-band spectrum that flattens reds and skin tones, while the incandescent emits a smooth, continuous spectrum close to true blackbody radiation. CCT alone can’t tell you this — you need to look at the full spectrum, or at the fidelity and gamut metrics derived from it.

The short version of each metric family

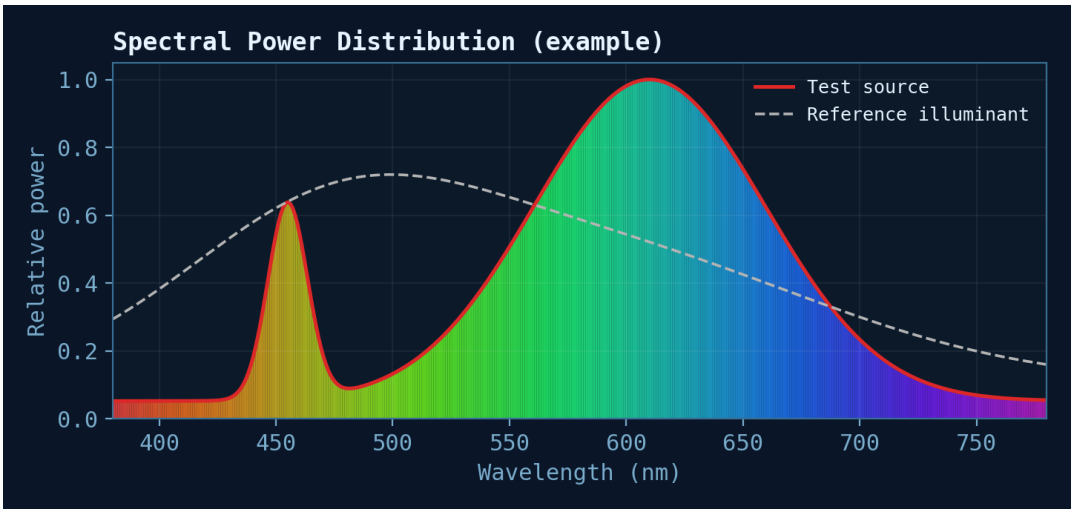
- **CRI (Ra, 0–100)** — the traditional 1974 CIE method, based on 8 pastel test colors. Ra is an average; the little-used R9 (a saturated red) often reveals problems Ra hides.
- **TM-30 (Rf, Rg)** — a modern replacement developed by the IES, using 99 real-world color samples. Rf plays a similar role to Ra but is more robust; Rg measures whether colors look more or less saturated than under a reference light, something CRI cannot capture at all.
- **CCT & Duv** — CCT is the closest blackbody temperature match; Duv is how far above or below the blackbody curve the light actually sits (its tint, independent of warmth).
- **LER** — how many lumens a light makes per watt of optical power. It says nothing about color quality, but it shows the efficiency price of a spectrum: pushing energy into deep red or far blue buys color at the cost of lumens.

hCRI.io computes all of these from the same underlying spectral measurement, so you’re never comparing numbers that came from different methodologies or assumptions.

3. Core concepts & glossary

SPD — Spectral Power Distribution

The raw measurement: how much light is emitted at each wavelength from 380 to 780nm. Every other metric on the page is derived from this curve. The colored fill under the curve represents the hue of each wavelength; the dashed grey line is the reference illuminant used for comparison.



Example SPD chart. The red line is the measured source; the grey dashed line is the reference.

CCT — Correlated Color Temperature (Kelvin)

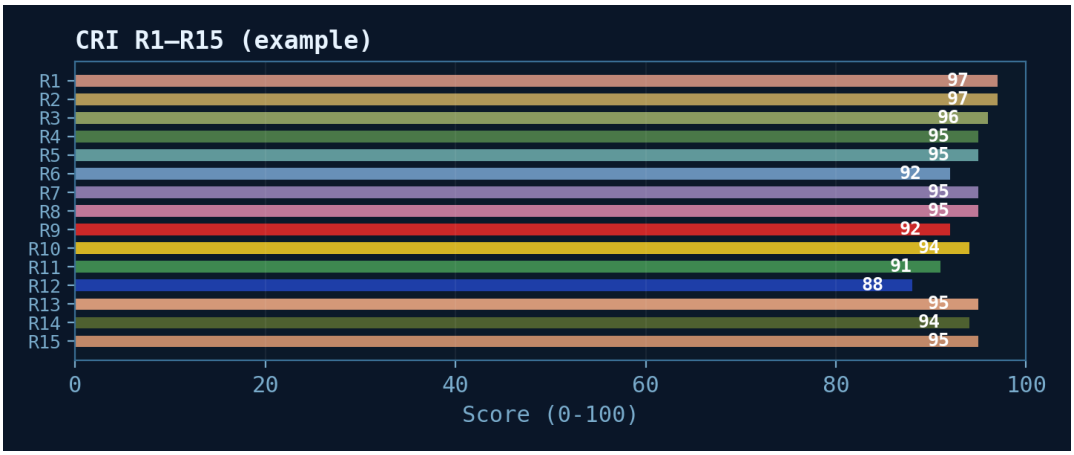
How warm or cool the light appears, expressed as the temperature of the closest-matching blackbody radiator. Lower numbers (2200–3000K) look warm/amber; higher numbers (5000–6500K+) look cool/blue.

Duv — Distance from the blackbody locus

CCT alone doesn't fully describe a light's tint. Duv measures how far above (greenish) or below (pinkish/magenta) the pure blackbody curve the source actually sits. A Duv near 0.0000 means the light tracks blackbody radiation closely; larger magnitudes mean a visible green or magenta cast.

Ra / CRI (CIE 13.3-1995)

The classic Color Rendering Index. Ra averages how accurately 8 standard pastel test colors (R1–R8) are reproduced, each scored 0–100. R9–R15 extend the set with a saturated red, yellow, green, blue, and skin-tone/foliage references — R9 in particular is a common weak point for inexpensive LEDs that a high Ra score can otherwise hide.



Example CRI R1–R15 breakdown. R9 (saturated red) is a common weak point.

Rf — TM-30 Fidelity

The IES TM-30-18 fidelity index, 0–100, computed from 99 real-world color samples rather than CRI's 8 pastels. Higher is more accurate. Because it draws on a much larger and more diverse sample set, Rf is considered a more reliable general-purpose accuracy score than Ra.

Rg — TM-30 Gamut

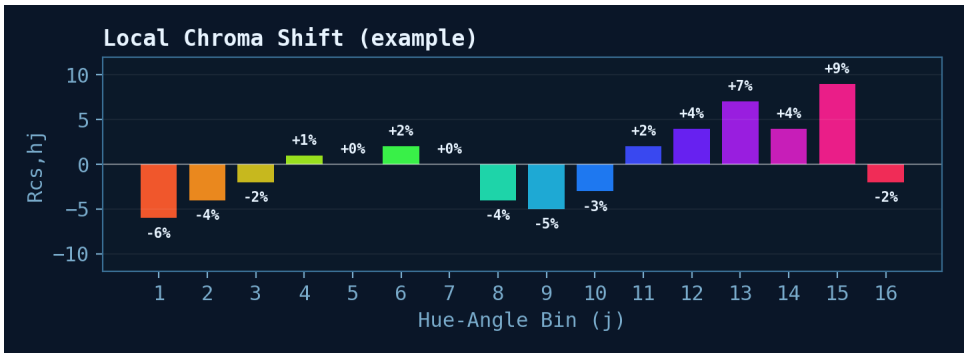
Measures whether colors appear more saturated (Rg > 100) or less saturated (Rg < 100) than under the reference illuminant, at the same fidelity. This is something CRI has no equivalent for at all — two lights with identical Ra can still make colors look noticeably more or less vivid.

Local Color Fidelity, Chroma Shift, and Hue Shift (Rf,hj / Rcs,hj / Rhs,hj)

TM-30 divides the color wheel into 16 hue-angle bins and reports fidelity, chroma shift, and hue shift for each one. This is where you find out which colors are affected, not just an average — e.g. a light might render greens and blues perfectly while pushing reds toward orange.



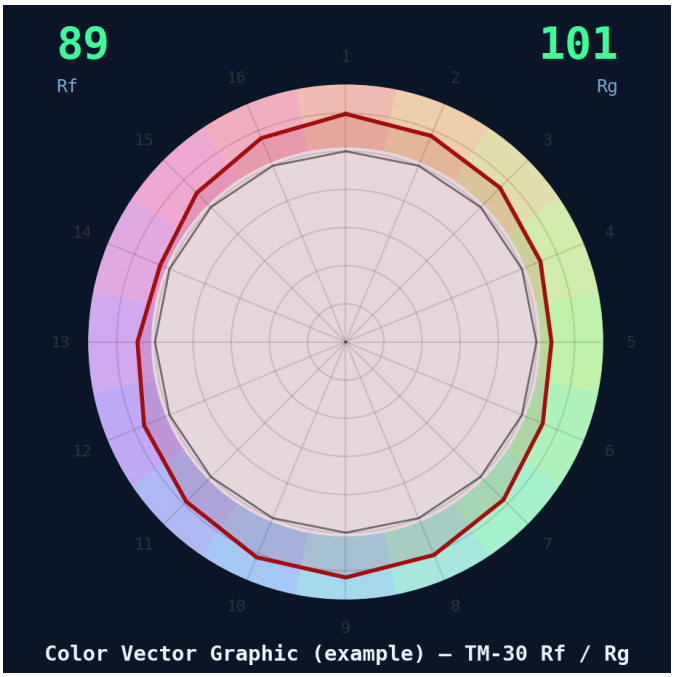
Local Color Fidelity by hue bin — shows exactly which hues are weakest.



Local Chroma Shift by hue bin — positive means more saturated than reference, negative means less.

Color Vector Graphic (CVG)

A polar plot showing how each of the 16 hue bins shifts in saturation and hue relative to the reference illuminant. Points pushed outward from the circle are rendered more saturated; points pulled inward are less saturated. The overall shape of the red trace is a quick visual signature of how a light source behaves.



Example Color Vector Graphic (CVG) with Rf and Rg overlaid.

CIE 1931 Chromaticity

The classic horseshoe-shaped diagram outlining every color the human eye can perceive. The curved line running through the middle is the Planckian (blackbody) locus — the path traced by an ideal blackbody radiator as its temperature changes. A light's (x, y) coordinate is plotted as a single point on this diagram; Duv is that point's distance from the locus.

SDCM — Standard Deviation of Color Matching

A tighter, practical companion to the CIE diagrams: ellipses ("MacAdam ellipses") around a target point showing how far a color point can drift before the difference becomes visible to the average human eye — useful for judging batch-to-batch consistency between fixtures.

LER — Luminous Efficacy of Radiation

Lumens per optical watt: 683.002 x (integral of V(lambda) S(lambda)) / (integral of S(lambda)), over 380-780 nm. V(lambda) is the eye's daylight sensitivity curve and S(lambda) is your measured spectrum. A warm-white phosphor LED is typically 280-340 lm/W; a spectrum with a lot of deep red or far blue scores lower. If your instrument did not measure the whole 380-780 nm range, the value uses only the measured part and is marked with an asterisk (*). The LER chart card draws your spectrum over V(lambda) and can also overlay the CIE S 026 alpha-opic curves (S, M and L cones, rods and melanopic).

Supported formats

- **CSV / TSV / TXT** — two columns, wavelength (nm) then intensity. Instrument metadata in header rows (CCT, Ra, R9, R1–R15, etc.) is picked up automatically when present.
- **JSON** — a "spd" object mapping wavelength to value, or a "metrics" object with pre-computed cct/duv/ra/r9.
- **Argyll CMS .sp** — standard CGATS spectral format, e.g. from a ColorMunki read with Argyll.
- **PDF** — some spectrometer measurement reports can be parsed directly for their SPD or summary metrics.

Three ways to upload

- **Try it free (no account)** — drop a file on the homepage for an instant report. It is not saved anywhere; create an account afterward if you want to keep it.
- **Signed in** — drag & drop or click to upload — supports multiple files and ZIP archives (extracted automatically) for batch uploads. Reports are saved to My Reports.
- **Paste data instead** — for quickly pasting a two-column list without saving a file first.

What new uploads start with

Every new report takes your account defaults, set in Profile → Privacy, and you can change any one report afterward from the report page:

- **Explore visibility** — public (visible on Explore) unless you turned on "Exclude my new reports from Explore by default".
- **Raw data download** — **on** by default (signed-in viewers can download the raw spectrum), unless you turn it off.
- **License** — **all rights reserved** by default, unless you pick another license as your default.

These defaults also apply to reports uploaded from the Companion app and through the API.

5. Reading a report, section by section

Every report page follows the same structure. Here's what each part shows and how to read it.

Header actions

- **TM-30** — opens the full IES TM-30-18 report page as a downloadable PNG or PDF, suitable for spec sheets.
- **Share Link / Copy Share Link** — generates a private link anyone can open, without needing an hCRI.io account.
- **Share Card** — a compact, shareable image card summarizing the key numbers, styled for social sharing or quick messaging.

Source details

The title, optional notes, and categorization tags (light brand/model, LED brand/model/CCT, optic) live here. Click any field to rename or tag it — these are also what's searchable and filterable across My Reports and Explore.

Sharing & data (your reports)

On your own reports a folded bar called **Sharing & data** holds three controls: **Exclude from Explore**, **Allow raw data download** and **License**. Expand it to change them; the change is saved immediately. On other people's reports you instead see their license in a one-line note near the title.

The metric grid

A row of key numbers — CCT, Duv, Ra, R9, Rf, Rg, LER and CIE x/y by default. This grid is customizable: drag cards to reorder them, remove ones you don't need, or add any raw field your instrument reported.

Color rendering charts

- **Color Vector Graphic** — the 16-bin polar plot described above, with Rf/Rg, CCT and Duv overlaid at center.
- **CRI R1–R15** — the classic per-sample bar chart.
- **CIE 1931 Chromaticity** — the source's (x, y) point plotted against the visible-color horseshoe and the blackbody locus.
- **Spectral Power Distribution** — the raw measured spectrum against the reference illuminant.
- **LER** — your spectrum against the eye's V(lambda) curve, with toggles for the alpha-opic curves.
- **Local Chroma Shift, Local Hue Shift, Local Color Fidelity** — the three 16-bin TM-30 breakdowns.
- **Color Sample Fidelity (99 samples)** — the same fidelity idea as the 16-bin chart, but at full per-sample resolution, useful for spotting a single problem color that a hue bin average might smooth over.

Every chart has a small "?" icon in its corner — hover or tap it for a short plain-language explanation without leaving the page. Charts and metric boxes can be dragged to reorder.

For more depth than a chart tooltip covers, the full Help & Reference library (opened from the help icon in the header, or at hcri.io/help/) has an article for every metric and feature, including this guide. Topics that mention each other are cross-linked automatically, and every article has its own shareable direct link via the Copy Link button, so you can point a colleague straight at the one topic they need.

Analysis

A short, automatically generated plain-language summary of the light's white point, fidelity, saturation, deep-red performance, and per-hue shifts, along with suggested uses (e.g. good for retail vs. better suited to ambient/accent

lighting).

See it on a photo & Raw Instrument Data

“See it on a photo” is covered in Section 8. At the bottom of the report, the **Raw Instrument Data** section holds the instrument headers and the full wavelength table, with a one-click CSV download. It appears when the owner’s **Allow raw data download** switch is on (the default) **and you are signed in**. Logged-out visitors, and share-link pages, never see it. Administrators always see the data in the admin report editor.

The first line of the CSV is `# License,<license> <link>` so the owner’s terms travel with the data. A downloaded spectrum cannot be recalled, so owners who do not want their data reused should switch the download off for that report (or for all reports, in Profile → Privacy).

Licenses

Each report carries a license chosen by its owner: **all rights reserved** (the default), CC0, CC BY, CC BY-SA, CC BY-NC, CC BY-ND, CC BY-NC-SA or CC BY-NC-ND (Creative Commons 4.0). Every license menu has a “What do these mean?” link to the manual (hcri.io/help/#report-licenses). A license states what others may do with the report and its data; it does not technically prevent copying, and it does not change the permission you give hCRI.io to display your reports (Terms of Service, section 4).

6. My Reports, Explore, Compare & Folders

My Reports

Every report you've uploaded, in a card grid or a compact single-column list (your choice, set per-page in your profile's Display settings). Organize reports into folders with drag-and-drop, filter by category (light brand, LED brand, CCT, etc.), and search by title.

Explore

A public gallery of reports other users have chosen to share. Great for browsing real-world examples across different light types before you have your own spectrometer data, or for comparing your results against similar fixtures. Opening a report from Explore shows its owner's license; the raw data section appears for signed-in visitors when the owner allows downloads.

Runtime curves

Explore → Runtime lists output-over-time curves (how a light's output holds up as it runs). Opening a comparison gives it its own address, so refreshing keeps it and Back returns to the list. "Share comparison link" creates a public link and confirms when it is copied.

Compare

Select two or more reports to view side by side, with a full-detail quick-view available for any card. Useful for A/B'ing two candidate fixtures, or checking whether a batch of "identical" LEDs actually measure consistently.

Folders

Group related reports — by project, by manufacturer, by installation — with simple drag-to-folder and drag-to-reorder organization. Folder context carries through to filtering, so you can narrow My Reports to just one folder at a time.

Filtering & Search

Narrow any report list by title search, category (light brand, LED brand, CCT, etc.), or by range sliders on CCT, Duv, Ra, R9, Rf, and Rg. Filters reset automatically when you switch between Explore, My Reports, and Folders, and the list scrolls back to the top whenever a filter changes or resets, so you're never left scrolled halfway down a list that's just been replaced.

Bulk Actions

Select multiple reports at once (up to 12) from My Reports to categorize or delete them together. Categorizing in bulk lets you apply a Light Brand, Light Model, LED CCT, LED Brand, LED Model, and/or Optic to every selected report in one step, with the option to overwrite existing values or only fill in categories that are still blank.

7. Sharing a report

hCRI.io gives you three separate sharing mechanisms, each suited to a different situation:

- **Public via Explore** — if a report is public, anyone can find it in the Explore gallery. Toggle this per-report, or set a default for all new uploads in your Privacy settings.
- **Share Link** — a private, unlisted link that works for anyone who has it, whether or not the report is public. Revoke it any time from the report page or from Profile → Links, which lists every active share link on your account.
- **Share Card** — a compact image summarizing Rf, Rg, CCT, Duv, Ra and R9, designed for dropping into a chat, email, or social post rather than linking to the full interactive report.

Share links are only created the moment you explicitly enable sharing — nothing is made shareable automatically just by uploading it. A shared report shows its license, but share-link pages never offer the raw data download, whatever the report's setting.

8. See it on a photo

Numbers are useful, but seeing the effect is more intuitive. “See it on a photo” takes a reference image and renders a side-by-side comparison: the left half shows the true, reference colors; the right half shows how those same colors would appear rendered under the measured light source, based on its per-hue chroma, hue, and lightness shifts.

- Use one of the two built-in sample photos, or drop in your own — photos are processed entirely in your browser and are never uploaded to any server.
- Drag the vertical divider to compare more or less of each side.
- The comparison updates live if you revisit the report after re-analyzing or updating instrument data.

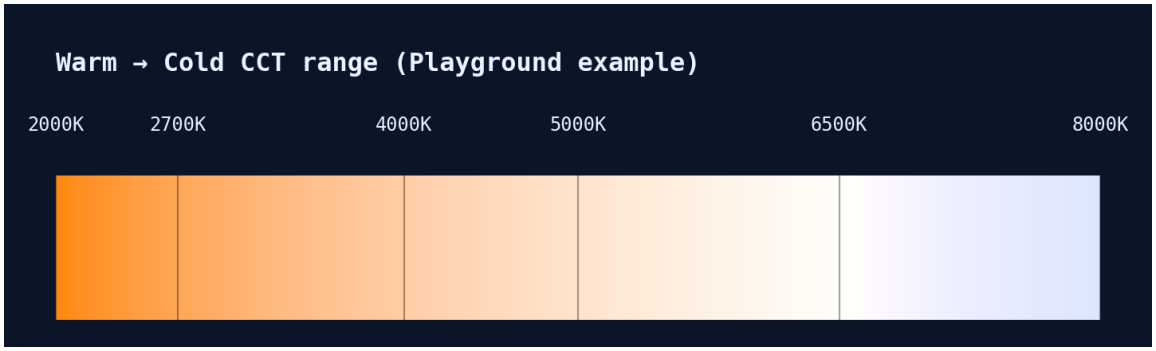
9. The Playground

The Playground is a standalone, interactive “what if” tool, separate from any specific uploaded report. Instead of analyzing a real measurement, you dial in fidelity, gamut, CCT, Duv, and individual R9/R10/R11/R12 values with sliders and watch the photo comparison update in real time.

Presets

A set of one-click presets is provided as starting points, from reference-grade sources through typical budget LEDs, plus a dedicated range of color-temperature presets running from very warm to very cold so you can see the visual effect of CCT and Duv in isolation from fidelity:

- **Halogen** and **High-CRI 95** — near-reference-quality sources.
- **Standard LED 80** and **Cheap LED 70** — typical and budget LED behavior, including the R9 (deep red) weakness common to inexpensive LEDs.
- **2000K Very Warm** through **8000K Very Cold** — six presets spanning the full warm-to-cold range at a constant quality level, isolating the visual effect of color temperature and tint on their own.



The Playground’s CCT presets span the full 2000K–8000K range, from very warm to very cold.

Use the Playground to build intuition for what a given Rf/Rg/CCT/Duv combination actually looks like before you go compare it against a real measurement in a report.

10. API access for automated uploads

If you're capturing measurements from a script, a lab instrument, or an automated test rig, hCRI.io supports non-expiring API tokens for uploading reports without going through the web UI. The full endpoint reference is a separate document, the hCRI.io API reference (hcri.io/assets/hcri-api-spec.pdf).

Creating a token

Profile → API → Create token. Treat a token like a password: it carries your account's upload access. Tokens can be named (e.g. "Lab laptop") and revoked individually at any time.

Uploading via curl

```
curl -X POST "https://www.hcri.io/index.php/api/v1/upload" \
  -H "Authorization: Bearer YOUR_TOKEN" \
  -F "file=@measurement.csv" \
  -F "label=My Light"
```

Accepts CSV, TSV, TXT, JSON, .sp, and PDF, the same formats as the web uploader. The response includes the same computed metrics you'd see on the report page, plus the report's `license`. A report created through the API takes your account defaults (Explore visibility, raw data download, license); owners change them afterward in the app or on the website. A token can only upload and annotate reports you own; it cannot edit account settings or read other people's data.

11. Account, privacy & data

Profile → Privacy is organized into collapsible sections. Each shows its current setting when closed, and the first three have a button that applies your choice to all of your existing reports after a confirmation.

- **Explore visibility** — choose whether new uploads are public or private by default; change any individual report afterward regardless of the default. “Apply to all” makes every report you own public or private at once.
- **Raw data download** — the default for new reports (on unless you change it). Per-report switches override it, and “Apply to all” sets every report at once. Only signed-in viewers can ever download.
- **License for my reports** — the default license for new reports, with “Apply to all” for existing ones.
- **Your data** — **Export** everything hCRI.io holds about your account as a single JSON file; or **delete your account**, which permanently erases your private reports and personal data. You can optionally keep your public Explore reports with your name removed, or delete everything.
- **Display preferences** — choose card-grid or single-list view independently for the Explore and My Reports pages.
- **Timezone** — every date and time on the site (report timestamps, activity logs, etc.) defaults to Eastern Time. Set your own from Profile → Display if you’d rather see everything in your local timezone instead.

Site announcements

From time to time a banner appears at the top of the site or on report pages (an announcement, a site issue, a warning). Close it with the X; it stays closed on that browser until the message is edited or a new one is posted. Banners may contain links.

12. Quick-start: your first report in 5 steps

1. Get a spectral measurement

Export an SPD from your spectrometer as CSV, TSV, JSON, or Argyll .sp — or use a PDF export if your instrument produces one.

2. Upload it

On hcri.io, drag the file onto the upload area (no account needed to try it), or sign in first if you want the report saved to My Reports.

3. Read the headline numbers

Start with Rf (fidelity) and Rg (gamut) at the top, alongside CCT and Duv. These four numbers give you the overall picture in seconds.

4. Drill into the hue-bin charts

If something looks off, the Local Color Fidelity, Chroma Shift, and Hue Shift charts — or the 99-sample fidelity chart — will show you exactly which colors are affected.

5. Share or save it

Rename and tag the report, choose its license, then use Share Link or Share Card to send it to a colleague or client, or download the full TM-30 report as a PDF for a spec sheet.

Appendix A: The Technology Behind hCRI.io

hCRI.io is built with a deliberately lean stack — no heavy frontend framework tooling beyond what's needed, and no backend framework at all.

Frontend

- React 19, built with Vite
- Three.js for the 3D CRI playground
- Canvas-based rendering for charts (SPD, CVG, CRI bars, chromaticity diagrams) rather than an external charting library
- Hand-rolled styling — no CSS framework; dark/light theme via a React context

Backend

- PHP, no framework — a single entry point handles routing
- PDO/MySQL for the database
- FPDF for generating downloadable TM-30 PDF reports
- PHPMailer for outgoing email (password resets, contact form)
- Apache with mod_rewrite for URL routing in production

Other

- Umami for cookieless, privacy-respecting analytics
- Argyll CMS .sp format support alongside CSV/TSV/JSON spectral data ingestion
- The hCRI Companion mobile app (React Native) for instruments and uploads

For the color-science implementation specifically, see Section 3 (Core concepts & glossary) and Appendix B below.

Appendix B: Glossary of Terms

Quick one-line definitions for reference. See Section 3 for the fuller explanations with examples.

Blackbody / Planckian radiator — An idealized light source whose color is determined purely by its temperature; the reference curve CCT and Duv are measured against.

CCT (Correlated Color Temperature) — How warm or cool a light appears, in Kelvin, based on the closest-matching blackbody temperature.

CES (Color Evaluation Samples) — The 99 real-world reflectance samples TM-30 uses to compute Rf and Rg, at both 16-bin and full 99-sample resolution.

CIE 1931 Chromaticity Diagram — The standard horseshoe-shaped diagram plotting every color the eye can perceive, with the Planckian locus running through it.

CRI / Ra — The classic 1974 CIE Color Rendering Index; Ra averages 8 pastel test-color scores (R1–R8), each 0–100.

CVG (Color Vector Graphic) — A polar plot showing how each of TM-30's 16 hue bins shifts in saturation and hue versus the reference illuminant.

Duv — A light's distance above (greenish) or below (pinkish) the Planckian locus — its tint, independent of CCT.

Hue-angle bin — One of 16 equal segments of the color wheel that TM-30's local metrics (Rf,hj / Rcs,hj / Rhs,hj) are reported per-segment for.

LER (Luminous Efficacy of Radiation) — Lumens per optical watt of a spectrum, computed from the eye's V(lambda) curve; unrelated to color quality.

License — The terms an owner attaches to a report: all rights reserved (default), CC0, or one of six Creative Commons 4.0 licenses.

MacAdam ellipse / SDCM — An ellipse around a target color point showing how far a color can drift before the difference becomes visible to the average eye.

R1–R15 — The individual CRI test-color scores; R1–R8 average to Ra, while R9–R15 add a saturated red, yellow, green, blue, skin tone, and foliage.

R9 — The saturated-red CRI test color, excluded from the Ra average — a common weak point for inexpensive LEDs.

Raw data download — The per-report switch (on by default) that lets signed-in viewers download the underlying spectrum as CSV.

Rcs,hj (Local Chroma Shift) — How much more or less saturated a given hue bin appears versus the reference illuminant.

Rf (TM-30 Fidelity) — The IES TM-30-18 fidelity score, 0–100, computed from 99 real-world color samples rather than CRI's 8 pastels.

Rf,hj (Local Color Fidelity) — TM-30's fidelity score reported separately for each of the 16 hue bins, rather than as one overall number.

Rg (TM-30 Gamut) — Whether colors appear more (>100) or less (<100) saturated overall than under the reference illuminant, at matched fidelity.

Rhs,hj (Local Hue Shift) — How many degrees a given hue bin rotates toward a neighboring hue versus the reference illuminant.

SPD (Spectral Power Distribution) — The raw measurement of how much light is emitted at each wavelength; the source data every other metric is derived from.

TM-30 (IES TM-30-18) — The modern IES color-rendition standard using 99 color samples, producing the Rf and Rg scores.

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